

loop

```
state = bicopter.Step(act_cmd)
```

```
 $\omega_{body} = state[\sim]$ 
```

```
act_cmd = fsw.get_act_cmd(throttle_frac,  $\omega_{des}$ ,  $\omega_{meas}$ )
```

fsw

↳ controller (object) needs prior data

↳ allocator (function) ~ need prior data

↳ vehicle geometry → config["geometric_properties"]

↳ throttle2force_lookup



```
fsw.get_act_cmd( $\omega_{des}$ ,  $\omega_{meas}$ , throttle_frac)
```

```
Mcmd = self.controller( $\omega_{des}$ ,  $\omega_{meas}$ )
```

```
act_cmd = self.allocator(Mcmd,  $f_z$ , vehicle_geometry)
```